Unit 4
Data and Analysis

SQLite · pandas · matplotlib · mean / median / spread

SQLITE3 DATAFRAME BAR · HIST · SCATTER · BOX $\bar{x}$, MEDIAN, σ

Schools, shops, and hospitals already decide from data. This unit is how a Python program joins that conversation.

0. Analysis is a pipeline, not a chart

Data analysis examines, organises, and interprets data so someone can decide. Python is popular because it reads files and databases, structures rows, computes statistics, and draws graphs — one language from source to slide.

source → connect → clean → compute → visualise → decide

1. Talking to a database from Python

A program cannot "see" a database until a **connection** exists. Stack: database engine → driver → connection → cursor → SQL → `commit()` / `fetchall()` → `close()`.

```
import sqlite3
con = sqlite3.connect("school.db")
cur = con.cursor()
cur.execute("""CREATE TABLE IF NOT EXISTS marks (
    id INTEGER PRIMARY KEY, name TEXT, math INTEGER, science INTEGER)""")
cur.execute("INSERT INTO marks (name, math, science) VALUES (?,?,?)",
            ("Ayesha", 88, 91))
con.commit()
cur.execute("SELECT name, math FROM marks WHERE math >= 80")
print(cur.fetchall())
con.close()
```

SQL verbs the course expects: `CREATE`, `INSERT`, `SELECT`, `UPDATE`, `DELETE`. SQLite is a single file — perfect for labs with no server.

2. pandas DataFrames — a table that can think

```
import pandas as pd
df = pd.read_csv("marks.csv")
# df = pd.read_excel("marks.xlsx", sheet_name="Term1")
df = pd.DataFrame({"Name": ["Ali","Sara"], "Math": [70, 90], "Science": [80, 85]})
df["Total"] = df["Math"] + df["Science"]
print(df.head())
print(df.isnull().sum())
df["Math"] = df["Math"].fillna(df["Math"].mean())
df = df.dropna()          # last resort: drop broken rows
```

Missing cells arrive as `NaN`. Filling with the column mean keeps n intact; dropping rows is honest when the row is junk. Never fill first and think later.

3. Organising for a human eye

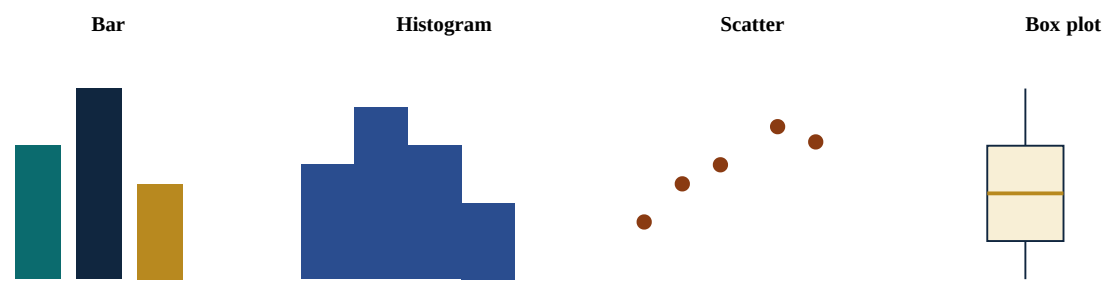
TABULAR
Rows and named columns — the DataFrame itself.

FREQUENCY TABLE
Score bands (0–9, 10–19, …) and how many students sit in each.

GRAPH
A picture of the same numbers. Choose the plot from the question, not from which function you remember.

4. matplotlib — six plots, six jobs

```
import matplotlib.pyplot as plt
plt.figure()
plt.plot(years, averages, marker="o")     # trend
plt.bar(subjects, marks)                   # compare categories
plt.pie(sizes, labels=labels, autopct="%1.1f%%")
plt.hist(scores, bins=8)                   # distribution; bars touch
plt.scatter(rain, yield_)                  # relationship
plt.boxplot([class_a, class_b])           # spread + outliers
plt.xlabel("..."); plt.ylabel("..."); plt.title("...")
plt.grid(True); plt.tight_layout(); plt.show()
```



Bar compares labels. Hist shows a continuous variable. Scatter asks “do X and Y move together?” Box shows median and spread.

Figure 1. Four of the six exam plots. Also practise line (time) and pie (parts of a whole).

Table 1. Pick the plot from the question

Question	Plot
How did yearly averages move?	Line
Which subject scored higher?	Bar
What share chose Biology?	Pie
How are scores distributed?	Histogram
Does yield rise with rain?	Scatter
Which class is more spread out?	Box plot

5. Descriptive statistics — the numbers behind the picture

For a list $x_1, x_2, \ldots, x_n$:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \text{Range} = x_{\max} - x_{\min}, \qquad \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad \sigma = \sqrt{\sigma^2}$$

(Some courses use $n - 1$ in the variance. Follow the mark scheme; the idea is “average squared distance from the mean”.)

- **Mean** uses every value — pulled by outliers (one huge salary).
- **Median** is the middle after sorting — safer for skewed pay.
- **Mode** is the most frequent — the right average for “favourite fruit”.

Table 2. Which centre?

Situation	Use	Why
Balanced test scores	Mean	Uses all information
Salaries with two tycoons	Median	Resists extremes
Favourite fruit / blood group	Mode	Category, not a number line

Spread: range (quick, fragile), variance / standard deviation (typical distance from $\bar{x}$), min and max (the edges), box-plot quartiles Q_1, median, Q_3.

Self-check

1. Marks: $10, 12, 13, 13, 80$. Mean vs median — which tells the truth about a "typical" student?
2. Why do histogram bars touch when bar-chart bars usually do not?
3. Write the pandas line that replaces missing Science marks with the column mean.

Answers. (1) Median 13; the mean is dragged toward 80. (2) A histogram bins a continuous axis; neighbouring bins share a boundary. (3) `df["Science"]=df["Science"].fillna(df["Science"].mean())`.

Recap

1. `sqlite3`: connect, cursor, execute, commit, fetch, close.
2. pandas: `read_csv`, new columns, `isnull`, `fillna`, `dropna`.
3. Six plots: line, bar, pie, hist, scatter, box — match plot to question.
4. Centre: mean / median / mode. Spread: range, variance, σ, quartiles.

Chart one CSV three ways: bar, hist, box. Write one sentence each chart allows.